LESSON 2.5

Describe Data by Range and Interquartile Range

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the range and the interquartile range of a data set and use them to describe how spread out the data is.

WHY THIS MATTERS The league analyst is comparing two sixth-grade classes that took the same math test. Both classes have the SAME median score, 81 — so the medians alone cannot tell them apart. Your job is to measure how spread out each class's scores are, using the range and the interquartile range, and decide which class did better overall.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Range and Interquartile Range *Spanish: Rango y rango intercuartílico*	Two numbers that describe spread: the range covers all the data, and the interquartile range covers only the middle half.	Range 26 covers 68 to 94; IQR 12.5 covers only 75.5 to 88
Range *Spanish: Rango*	The distance from the least value to the greatest value, found by subtracting: greatest − least.	greatest − least
Quartile *Spanish: Cuartil*	One of the three values that cut an ordered data set into four equal-sized groups.	Q1, median, Q3
Interquartile range *Spanish: Rango intercuartílico*	The distance across the middle half of the data, found by subtracting the first quartile from the third: Q3 − Q1.	IQR = Q3 − Q1
Variability *Spanish: Variabilidad*	How much the values in a data set differ from one another — whether they sit close together or far apart.	Close together or far apart
Measure of variation *Spanish: Medida de variación*	A single number that describes the spread of a whole data set. The range and the IQR are both measures of variation.	Range and IQR are both

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Hikes (km): 5, 12, 8, 20, 15, 10, 18.

1. Order: 5, 8, 10, 12, 15, 18, 20.
2. Range = 20 − 5 = 15 km.
3. Q1 = 8, Q3 = 18, so IQR = 18 − 8 = 10 km.

Answer: Range 15 km; IQR 10 km

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Class A's box plot reads: least 62, Q1 74, median 81, Q3 87, greatest 90. Find the range first.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The greatest value is _____ and the least value is _____ .
- The range is _____ , which describes _____ .
- Quartile 1 is _____ and quartile 3 is _____ , so the IQR is _____ .
- The IQR describes only _____ of the data.

Word bank range | interquartile range | quartile | spread | variability | greatest | least | subtract | middle half | extreme | order | consistent

WATCH OUT!

A common mistake

A common mistake with Range and Interquartile Range is adding the two quartiles instead of subtracting them

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Two classes both have a median score of 81. Class A's IQR is 13 and Class B's IQR is 8. Which statement is true?

(A) Class B has a higher median

(B) Class A has more students

(C) The middle half of Class B's scores is less spread out

(D) Class B's highest score is greater

STEPS TO TRY

1. A smaller IQR means a narrower box.
2. A narrower box means the middle half of the values sit closer together.
3. Which class has the smaller IQR?

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A data set runs from a least value of 12 to a greatest value of 30. What is the range?

(A) 12

(B) 18

(C) 21

(D) 30

STEPS TO TRY

1. Range asks how far it is from one end of the data to the other.
2. Find the greatest value and the least value, then subtract.
3. 30 − 12 = ?

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

3 **Circle the letter of the best answer. Show how you know.**

On a box plot, Q1 = 20 and Q3 = 36. What is the interquartile range?

(A) 16
(B) 28
(C) 36
(D) 56

STEPS TO TRY

1. The IQR is the width of the box on a box plot.
2. The box runs from Q1 to Q3, so subtract Q1 from Q3.
3. 36 − 20 = ?

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

4 **Circle the letter of the best answer. Show how you know.**

Which measure of spread uses ONLY the middle half of the data?

(A) The interquartile range
(B) The range
(C) The median
(D) The mode

STEPS TO TRY

1. One is measured across the box on a box plot; the other whisker to whisker.
2. The box holds the middle 50% of the data.
3. Q1 to Q3 is the middle half.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Both numbers describe spread. Why might a coach trust the IQR more than the range when one player had a record-breaking night?

SENTENCE STARTER I would use the ___ because it only uses the ___, so one unusual value cannot ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help